

ECE 684: Natural Language Processing
Assignment 6 – Latent Dirichlet Allocation

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Q. Show the inferred beta vectors and indicate how they map to the true topics above.

A. **Beta values:**

```
Beta values:
[[0.4 0.4 0.2 0. 0. 0. ]
 [0. 0.3 0.1 0. 0.3 0.3]
 [0.3 0. 0.2 0.3 0.2 0. ]]
```

Inferred Beta values:

```
[Running] python -u "/Users/nrutachoudhari/Documents/MIDS/Fall 2024/ECE 684 Natural Language Processing/Latent Dirichlet Allocation 0910/
tempCodeRunnerFile.py"
[0.33333334 0.33333334 0.33333334]
[(0, '0.279*tuba' + 0.224*horn' + 0.218*bass' + 0.205*deep' + 0.043*pike' + 0.032*catapult'), (1, '0.341*bass' + 0.230*pike' + 0.
187*deep' + 0.114*horn' + 0.087*tuba' + 0.041*catapult'), (2, '0.358*pike' + 0.204*horn' + 0.175*catapult' + 0.128*deep' + 0.119*bass"
+ 0.016*tuba'')]
```

Inferred Beta Vectors (Topics):

Topic 0: 0.279*'tuba' + 0.224*'horn' + 0.218*'bass' + 0.205*'deep' + 0.043*'pike' + 0.032*'catapult'

Topic 1: 0.341*'bass' + 0.230*'pike' + 0.187*'deep' + 0.114*'horn' + 0.087*'tuba' + 0.041*'catapult'

Topic 2: 0.358*'pike' + 0.204*'horn' + 0.175*'catapult' + 0.128*'deep' + 0.119*'bass' + 0.016*'tuba'

In the beta values:

Topic 1: [0.4 0.4 0.2 0.0 0.0 0.0]

Topic 2: [0.0 0.3 0.1 0.0 0.3 0.3]

Topic 3: [0.3 0.0 0.2 0.3 0.2 0.0]

From the inferred values:

Inferred Topic 0:

- “tuba” (0.279), “horn” (0.224) and “bass” (0.218) suggest a strong musical theme
- Other words like “deep” (0.205) may also relate to musical terms, indicating that this topic primarily focuses on music-related concepts
- This split is consistent with Topic 2

Inferred Topic 1:

- “bass” (0.341), “pike” (0.230) could suggest a theme related to fishing, as “bass” also refers to a type of fish.
- Words like “deep” (0.187) and “horn” (0.114) might also relate to sound or depth, adding complexity to the topic
- This split is consistent with Topic 1

Inferred Topic 2:

- “pike” (0.358), “horn” (0.204) and “catapult” (0.175) indicates a hunting theme.
- The split is consistent with Topic 3